KPPSC PMS • GEOGRAPHY (PAPER-II)

EXECUTIVE MASTER EXAMINATION NOTES • Q.17: Power Resources & Global Energy Geopolitics: Coal, Petroleum/Crude Oil, Gas & Alternative Energy

Target Examination	KPPSC PMS (Provincial Management Service) — Khyber Pakhtunkhwa
Paper & Subject	Geography Paper-II (Human, Economic & Regional Geography — 100 Marks)
EXAM QUESTIONS LINKED:	• 2016 Q.6: "\"The spatial distribution of power resources influences world geopolitics and economic supremacy.\" Discuss. (20 Marks)" • 2013 Q.3: "Discuss methods of conservation of raw material resources. Critically evaluate the potential for alternative energy sources. (20 Marks)" • 2010 Q.7: "\"Power resources are playing a key role in economic development.\" Describe the production and trade of Crude Oil in the world. (20 Marks)" • 2008 Q.7(iv): "Write short note on: Coal deposits of the USA. (10 Marks)"
Syllabus Section	Section B & C: Economic & Regional Geography — Power resources of the world and Pakistan: coal, petroleum, gas, renewables
Allocated Marks & Time	20 Marks | 35 Minutes Exam Writing Allocation
Core Analytical Thesis	Power resources constitute the primary energetic metabolic engine of modern industrial civilization. Global energy geography is structured along a historical transition from solid hydrocarbons (coal stratification: anthracite, bituminous, lignite) through liquid and gaseous hydrocarbons (petroleum geopolitics of the Persian Gulf and natural gas pipeline diplomacy) to decarbonized renewable energy systems (hydel, solar, wind, and nuclear). In Pakistan, the power sector is crippled by a profound geographic and financial contradiction: severe dependence on expensive imported fossil fuels (RLNG, furnace oil, coal) driving an unsustainable circular debt crisis (>PKR 2.6 trillion), juxtaposed against immense untapped domestic reserves: 175 billion tons of Thar lignite coal, 60,000 MW of Himalayan-Karakoram hydroelectric potential, and prolific solar/wind corridors across southern Sindh and Balochistan.

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📌 EXAM ESSENTIAL TAKEAWAY

HYDROCARBON THERMODYNAMIC LADDER: Coal evolves through heat and pressure: Peat (<60% carbon) ⟶ Lignite (60-70% carbon, brown coal, high moisture) ⟶ Bituminous (75-85% carbon, steam & coking coal) ⟶ Anthracite (>90% carbon, smokeless metamorphic coal).

PETROLEUM & GAS GEOPOLITICS: Crude oil and natural gas are concentrated in anticlinal structural traps across sedimentary basins; the Persian Gulf holds ~48% of global proven oil reserves, making the Strait of Hormuz the world's most critical maritime chokepoint (21M barrels/day).

THE ENERGY TRILEMMA: World Energy Council framework balancing: (1) Energy Security (reliable supply), (2) Energy Equity (affordable access), and (3) Environmental Sustainability (decarbonization / net zero).

> PAKISTAN'S POWER SECTOR PARADOX: Installed generation capacity has reached ~45,000 MW, yet the economy suffers from chronic load-shedding and extortionate tariffs due to flawed 'Take-or-Pay' Independent Power Producer (IPP) contracts, transmission bottlenecks (evacuation limit ~25,000 MW), and reliance on imported fuels.

1. CONCEPTUAL MIND MAP & THEMATIC ROADMAP

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GLOBAL ENERGY GEOGRAPHY & PAKISTAN POWER SECTOR DYNAMICS
=========================================================================================================
[1. THE GLOBAL ENERGY RESOURCE SPECTRUM]
  ├── (A) SOLID FOSSIL FUELS (Coal Ranks & Basins)
  │      ├── Bituminous & Anthracite: Appalachian Basin (USA), Ruhr/Silesia (Europe), Shanxi (China), Donbas.
  │      └── Lignite: Global deposits; Thar Coal Field (Sindh, Pakistan) — 175 Billion Tons (7th largest in world).
  ├── (B) LIQUID HYDROCARBONS (Petroleum & Geopolitical Chokepoints)
  │      ├── Reserve Giants: Persian Gulf (Saudi, Iran, Iraq, UAE), Venezuela (Orinoco Belt), Russia (West Siberia).
  │      └── Strategic Maritime Chokepoints: Strait of Hormuz, Malacca Strait, Bab el-Mandeb, Suez Canal.
  ├── (C) GASEOUS HYDROCARBONS (Natural Gas & LNG)
  │      ├── Massive Basins: South Pars / North Dome (Qatar/Iran), Yamal (Russia), Permian Basin (US Shale Gas).
  │      └── Transport Duality: Fixed Overland Pipelines vs. Cryogenic Liquefied Natural Gas (LNG) tankers.
  └── (D) RENEWABLE & ZERO-CARBON ALTERNATIVES
         ├── Hydropower: Grand Coulee, Three Gorges, Indus Cascade (Tarbela, Dasu, Diamer-Bhasha).
         ├── Nuclear Energy: Baseload low-carbon fission (PWR/BWR); France (~70%), China, Pakistan (Chashma, K2/K3).
         └── Solar & Wind: Desert photovoltaic belts (Sahara, Thar, Quetta) and coastal wind corridors (Gharo-Jhimpir).
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         ▼ (The Structural Breakdown of Pakistan's Power Matrix)
[2. PAKISTAN'S ENERGY MATRIX & FINANCIAL STRANGULATION]
  * Generation Energy Mix: Hydel (~30%) | Imported Coal (~10%) | Thar Domestic Coal (~12%) | RLNG/Gas (~25%) | Nuclear
(~12%) | Wind/Solar (~5%).
  * The Circular Debt Abyss: Exceeds PKR 2.6 Trillion (driven by capacity payments to IPPs, dollar-indexed tariffs).
  * Transmission Evacuation Cap: Grid capacity maxes out at ~24,000-25,000 MW despite ~45,000 MW installed capacity.
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         ▼ (The Indigenous Energy Pivot Strategy)
[3. THE THREE INDIGENOUS ENERGY CORRIDORS]
  ├── (1) Thar Coal Block Monetization: Mine-mouth power plants (Engro, Shanghai Electric); coal-to-gas/liquid
potential.
  ├── (2) The Indus River Hydel Cascade: Dasu (4,320 MW), Diamer-Bhasha (4,500 MW), Mohmand Dam (800 MW).
  └── (3) Wind & Solar Corridors: Gharo-Jhimpir wind belt (Sindh) with 50,000 MW potential; Balochistan solar
insolation.
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2. ANALYTICAL DIMENSIONS & SOCIO-SPATIAL/THEORETICAL MATRIX

Power Resource Type	Energy Density & Thermodynamic Characteristics	Global Distribution & Major Producers	Pakistan Spatial Footprint & Current Utilization
Coal (Anthracite & Bituminous)	High energy density (24–32 MJ/kg); high carbon emissions; heavy sulfur dioxide and ash byproduct.	China (world leader, >50% consumption), India, USA, Australia, Russia, Indonesia.	Domestic mining in Salt Range (Dandot, Makerwal), Quetta (Mach, Degari); largely used in brick kilns and cement.
Lignite (Brown Coal)	Low energy density (14–18 MJ/kg); high moisture content (40–50%); must be combusted at mine-mouth.	Germany, Russia, Australia, USA, Poland, Turkey.	Thar Coalfield (Sindh): 175 billion tons reserve (~9,000 km²); multiple 660 MW mine-mouth thermal plants operational under CPEC.
Petroleum / Crude Oil	High energy density (42–46 MJ/kg); liquid transport flexibility; primary transportation fuel.	USA, Saudi Arabia, Russia, Canada, Iraq, UAE, Iran. Persian Gulf contains 48% of global reserves.	Domestic production covers only ~15% of demand (Potohar: Khaur, Dhurnal; Lower Sindh: Badin); $15B+ annual import bill.
Natural Gas & LNG	Cleaner combustion (50–55 MJ/kg); lower CO_2 than coal; pipeline or cryogenic LNG shipping.	USA (shale gas), Russia, Qatar, Iran, Turkmenistan, Australia.	Historic Sui Gas Field (Balochistan, discovered 1952) severely depleted; massive import of expensive RLNG via Port Qasim.

Power Resource Type	Energy Density & Thermodynamic Characteristics	Global Distribution & Major Producers	Pakistan Spatial Footprint & Current Utilization
Hydroelectric Power	Zero fuel cost; high upfront capital; renewable; multipurpose (flood control & irrigation).	China (Three Gorges 22.5 GW), Brazil/Paraguay (Itaipu), Canada, USA, Norway (98% hydro).	Generates ~10,000 MW; Tarbela (4,888 MW), Mangla (1,000 MW), Ghazi-Barotha (1,450 MW); Dasu & Diamer-Bhasha under construction.
Nuclear Power	Hyper-dense baseload (uranium fission); zero greenhouse gas emissions; radioactive waste dilemma.	USA (93 reactors), France (56 reactors, 70% power), China (fastest growing), Japan, Russia.	PAEC operates 6 commercial reactors generating ~3,530 MW: Chashma (C1–C4) in Punjab, and K-2/K-3 (1,100 MW each) in Karachi.
Solar & Wind Renewables	Intermittent; zero fuel cost; rapidly falling solar PV panel capital costs; dependent on weather.	China (global titan in solar PV & wind turbine manufacturing), USA, Germany, India, Spain.	Gharo-Jhimpir wind corridor in Sindh (~1,800 MW installed); Quaid-e-Azam Solar Park (Bahawalpur); rapid net-metering adoption.

3. 20-MARK MASTER MODEL ESSAY (KPPSC PMS BENCHMARK)

Introduction: Energy as the Master Resource of Terrestrial Civilization

Energy is the fundamental currency of the physical universe and the indispensable metabolic engine of human society. In economic and regional geography, power resources dictate the location of industries, the velocity of transportation, the productivity of agriculture, and the standard of human living. The modern world was forged through successive energy transitions: from biomass and muscle power to coal in the nineteenth century, to petroleum and natural gas in the twentieth century, and currently toward a complex transition to renewable and nuclear power in the twenty-first century. For developing nations, energy is the supreme prerequisite for economic takeoff; without reliable, affordable power, factories close, foreign capital flees, and macroeconomic stability collapses. Pakistan's contemporary economic paralysis is, at its root, an energy geography crisis.

The Hydrocarbon Spectrum: Coal Thermodynamics and Petroleum Geopolitics

Fossil hydrocarbons remain the dominant primary energy source on Earth, accounting for roughly 80% of global primary energy consumption: - The Coal Spectrum and Thermodynamic Stratification: Coal is a combustible sedimentary rock formed from accumulated plant matter preserved in anaerobic swamp environments during the Carboniferous period (300 to 360 million years ago) and subjected to millions of years of geothermal heat and tectonic pressure. Coal evolves along a strict metamorphic ladder: (1) Peat (pre-coal, high moisture, low carbon <60%); (2) Lignite ('brown coal', low calorific value 14–18 MJ/kg, high moisture 40–50%, prone to spontaneous combustion, making long-distance transport uneconomic and requiring mine-mouth power plants); (3) Sub-bituminous and Bituminous Coal ('black coal', calorific value 24–30 MJ/kg, the global workhorse for electricity generation and steel smelting via metallurgical coking); and (4) Anthracite (metamorphic coal, hard and glossy, with over 90% fixed carbon, burning with an intense, smokeless flame). Leading coal basins—such as the Shanxi and Inner Mongolia basins in China, the Powder River Basin in the United States, and the Ruhr in Germany—historically determined the world's great manufacturing belts. - Petroleum and the Architecture of Global Geopolitics: Crude oil is a complex mixture of liquid hydrocarbons formed from ancient marine microscopic plankton buried in marine sedimentary basins. Oil migrates through permeable reservoir rocks (sandstones, porous limestones) until trapped by impermeable caprocks (shales, salt domes) in anticlinal folds or fault traps. The spatial geography of petroleum is acutely asymmetric: the Persian Gulf basin alone holds nearly half of all proved global conventional oil reserves. Consequently, global geopolitical security is inextricably tied to the defense of narrow maritime 'chokepoints' through which global oil tankers must pass—most critically the Strait of Hormuz (transiting 21 million barrels per day, roughly 21% of global petroleum consumption), the Strait of Malacca, the Bab el-Mandeb, and the Suez Canal. Disruptions at these chokepoints immediately transmit inflationary price shocks across the global economy. - Natural Gas and Pipeline Diplomacy: Natural gas (predominantly methane) is the cleanest burning fossil fuel, emitting roughly 50% less carbon dioxide per unit of energy than coal. The geography of natural gas is defined by transport mechanics: unlike oil, which is easily shipped in standard tankers, gas

must be transported either through high-pressure trans-continental overland pipelines (e.g., the Nord Stream or Central Asia-China pipelines) or chilled to -162°C to convert it into Liquefied Natural Gas (LNG) for specialized cryogenic maritime transport, linking major producers like Qatar, the United States, and Australia to energy-hungry consumers in Asia and Europe.

The Global Energy Transition and the 'Energy Trilemma'

In the twenty-first century, global energy geography is being reshaped by the 'Energy Trilemma' formulated by the World Energy Council, which requires states to balance three competing objectives: (1) Energy Security (reliable supply from diversified sources); (2) Energy Equity (universal access at affordable tariffs); and (3) Environmental Sustainability (decarbonization to prevent catastrophic global warming). Driven by climate imperatives and the plummeting costs of photovoltaic cells and wind turbines, the world is shifting toward clean energy: - Hydroelectric Power: Provides reliable, low-carbon, dispatchable baseload electricity, pioneered by massive engineering projects like the Three Gorges Dam in China (22,500 MW) and Itaipu on the Brazil-Paraguay border. - Nuclear Energy: Exploits the fission of Uranium-235 to generate massive thermal power with zero carbon emissions. Nations like France derive over 70% of their electricity from nuclear energy, demonstrating its viability as a safe baseload alternative to fossil fuels. - Solar Photovoltaics and Wind Turbines: The fastest-growing power sources globally, driven by technological breakthroughs that have slashed generation costs by more than 85% over the past decade, transforming desert sun-belts and coastal wind corridors into dynamic energy exporters.

The Anatomy of Pakistan's Power Sector Crisis: The Circular Debt Trap

Pakistan's power sector represents a classic geographic tragedy: a nation endowed with vast natural energy endowments that has engineered an artificial, crippling energy crisis through disastrous policy choices. The country's total installed power generation capacity has reached approximately 45,000 MW—theoretically more than sufficient to meet peak national summer demand (~30,000 MW). Yet, the nation suffers from rolling blackouts, uncompetitive industrial tariffs, and economic stagnation. This crisis is driven by three structural pathologies: 1. Flawed Contractual Architecture and 'Capacity Payments': During the 1990s and through the CPEC phase post-2015, the state attracted private investors by signing long-term (25 to 30 year) Independent Power Producer (IPP) contracts based on 'Take-or-Pay' dollar-indexed terms. Under these contracts, the government is legally obligated to pay massive 'capacity charges' to plant owners merely for keeping their plants available, regardless of whether a single unit of electricity is generated or consumed. In 2024, national capacity payments exceeded PKR 2.1 trillion, inflating electricity tariffs to over PKR 65 per kilowatt-hour, bankrupting commercial factories and household consumers. 2. Transmission Grid Bottlenecks: While generation capacity was expanded to 45,000 MW, the national transmission and distribution grid was never upgraded. The national grid can safely evacuate and transmit only 24,000 to 25,000 MW without tripping. Consequently, modern power plants sit idle while consumers face load-shedding, and the government continues to pay capacity charges for un-transmitted power. 3. The Import-Dependency Curse: Historically, Pakistan relied heavily on domestic natural gas (from the legendary Sui gas field in Balochistan, discovered in 1952) and domestic hydropower (Tarbela and Mangla). However, as domestic gas reserves were rapidly depleted due to reckless under-pricing for compressed natural gas (CNG) and fertilizer subsidies, Pakistan pivoted to expensive imported fossil fuels: imported furnace oil, imported coal (from South Africa and Indonesia for plants at Sahiwal and Port Qasim), and imported Regasified Liquefied Natural Gas (RLNG) from Qatar. When the Russia-Ukraine war sent international LNG and coal prices soaring in 2022, Pakistan's energy import bill surged to over $18 billion, exhausting central bank foreign exchange reserves, devaluing the rupee, and inflating the energy sector's 'Circular Debt' to an astronomical PKR 2.6 trillion.

Unlocking Pakistan's Domestic Energy Potential: Thar Coal, Indus Hydel, and Renewables

To escape this structural death spiral, Pakistan must re-anchor its power matrix to its indigenous geographical resources: - The Thar Lignite Coalfield: Located in the Tharparkar Desert of southeastern Sindh, Thar Coal represents one of the largest contiguous lignite coal deposits on Earth, containing an estimated 175 billion tons of coal across 9,000 km²—equivalent in energy content to the combined oil reserves of Saudi Arabia and Iran. Successfully developed under CPEC, several 660 MW mine-mouth thermal power plants (such as the Engro Thar and Shanghai Electric projects) are now operational, feeding thousands of megawatts of baseload power into the national grid at significantly lower fuel costs than imported coal. While burning lignite releases carbon emissions, utilizing Thar coal saves billions of dollars in foreign exchange. Furthermore, long-term spatial plans envision converting Thar coal into synthetic natural gas (coal-

to-gas) and urea fertilizer, slashing import bills. - The Indus River Hydroelectric Cascade: The northern gorge of the Indus River—where the river carves through the convergence of the Himalayas, Karakorams, and Hindu Kush—possesses over 60,000 MW of economically exploitable hydroelectric potential. Pakistan is currently constructing two mega-dams: Dasu Hydropower Project (4,320 MW) in KP's Kohistan district, and the monumental Diamer-Bhasha Dam (4,500 MW) on the border of KP and Gilgit-Baltistan. These projects will provide cheap, clean, renewable baseload electricity while storing 6.4 million acre-feet of water to secure national irrigation against catastrophic drought. - The Southern Wind and Solar Corridors: Southern Sindh hosts the world-class Gharo-Jhimpir wind corridor, with an exploitable wind power potential exceeding 50,000 MW within 100 km of Karachi. Simultaneously, Balochistan possesses some of the highest solar insolation rates in the world (>5.5 to 6.0 kWh/m²/day). Private citizens have already initiated an organic energy revolution, importing over $1.5 billion worth of cheap Chinese solar panels in 2023–24, installing thousands of megawatts of rooftop decentralized solar power across Punjab and KP.

Conclusion: Strategic Imperatives for National Energy Security

In conclusion, power resources represent the foundational physical substrate upon which modern national sovereignty and economic prosperity depend. The global transition away from carbon-intensive fossil fuels toward decentralized, zero-carbon renewable and nuclear power is accelerating. For Pakistan, resolving its chronic energy crisis requires an immediate, ruthless realignment with geographical reality: (1) renegotiate extortionate, dollar-indexed IPP capacity payment contracts to bring tariffs back to regional parity; (2) aggressively invest capital into upgrading the national transmission grid to eliminate evacuation bottlenecks; (3) halt all new imported-fuel power generation, substituting imported coal with domestic Thar lignite; and (4) rapidly complete the Dasu, Diamer-Bhasha, and Mohmand hydropower dams while modernizing net-metering regulations to integrate rooftop solar and coastal wind into a decentralized, resilient national smart grid.

4. SCHOLARLY PERSPECTIVES & GEOGRAPHICAL AXIOMS TABLE

Scholar / Geographer	School of Thought / Paradigm	Quotable Philosophical Thesis / Axiom
Vaclav Smil (2017)	Energy and Civilization: A History	*"Energy is the only universal currency; one of its many forms must be transformed to get anything done in the physical universe, and human civilization is fundamentally a story of mastering energetic flows."*
Daniel Yergin (1991)	The Prize: The Epic Quest for Oil, Money & Power	*"Petroleum is the blood of the modern industrial machine; control over oil reserves and maritime chokepoints has determined the outcome of world wars and the architecture of global hegemony."*
World Energy Council (2019)	The Energy Trilemma Index	*"True energy sustainability requires a balanced equilibrium among three pillars: security of supply, social equity through affordable access, and ecological decarbonization."*
Michael T. Klare (2001)	Resource Wars: The New Landscape of Global Conflict	*"The geography of twenty-first-century conflict will be written across energy basins and transit corridors; access to hydrocarbons and uranium remains the hidden engine of military strategy."*
National Electric Power Regulatory Authority (NEPRA)	State of the Industry Report (2023)	*"Pakistan's power sector suffers not from an absence of generation, but from an economic design failure: dollar-pegged capacity contracts and imported fuel dependence have created an unsustainable circular debt crisis."*